Linux Interview Questions

A categorized reference of Linux fundamentals, administration, troubleshooting, and networking questions

15 sections | **141** questions

Linux Fundamentals

1 What is an operating system?
2 Does the operating system communicate directly with the hardware?
3 What acts as a mediator between the operating system and hardware?
4 What are the different types of kernels?
5 Which operating system are you more familiar with—Linux or Windows?
6 Does Windows also have a kernel like Linux?

Linux Administration

1 What Linux distribution did you use?
2 What experience do you have with Linux/Unix systems and EC2 instances?
3 Can you give an example of work you performed on a Linux server?
4 How do you automate patching?
5 How do you patch Linux servers with minimal downtime?
6 Explain N-1 patching.
7 Patch lifecycle.
8 How do you validate after patch?
9 How do you manage kernel upgrades?
10 What is SELinux and how do you troubleshoot it?
11 How do you configure firewalld?
12 iptables vs firewalld?
13 How do you secure SSH access?
14 How do you configure static IP?
15 How do you configure time sync (NTP/chrony)?
16 How do you configure auditd?
17 How do you manage users and permissions?
18 How do you automate with Ansible?
19 What is a Bash script and how do you use it?

Linux Performance Troubleshooting

1. How do you check system resource utilization?
2. How do you debug high CPU usage?
3. How do you find high memory consuming processes?
4. Day-to-day Linux performance troubleshooting.
5. CPU is idle but load average is high.
6. Difference between CPU utilization and load average.
7. How do you dig deeper into an IO bottleneck?
8. Why are processes queued?
9. CPU 100%.
10. High IO wait.
11. High response time.
12. OOM Killer.
13. High CPU.
14. Memory leak.
15. High memory usage.

Memory & Swap

1. Explain the difference between RAM and Swap.
2. What is vm.swappiness?
3. Is high swap usage good or bad?
4. What triggers Out of Memory?
5. What happens when Linux OOM Killer is invoked?
6. What is swap memory?
7. What is the purpose of swap?
8. Is swap size always twice RAM?
9. Can an application requiring 12GB RAM run on a 6GB machine?
10. How does swap work?

Disk Management

1. How do you troubleshoot disk space issues?
2. Disk full.
3. inode full.
4. How do you check disk usage?
5. How do you check disk reads/writes?

6 Reasons for high disk reads/writes.
7 How do you troubleshoot a full disk?
8 How do you find directories consuming disk space?
9 List top 10 largest files/directories.
10 Root filesystem is 99.5% full.
11 /var/log consuming disk.
12 Application continuously generates logs.
13 Compress log files.
14 When do you increase disk size?
15 /boot partition full.
16 Extend LVM partitions.

File System

1 What is an inode?
2 Do directories have inode numbers?
3 Can file and directory have same name?
4 Default file permissions.
5 Default directory permissions.
6 What is umask?
7 Can you cd into a directory with 600 permissions?
8 How do you give full permissions to a file?
9 What are different file types?
10 Soft link vs Hard link.

Mounting

1 How do you mount a filesystem?
2 How do you unmount a filesystem?
3 Will mounted filesystem persist after reboot?
4 How do you make mount persistent?
5 How do you mount NFS?

Process Management

1 What is a zombie process?
2 How do you identify zombie processes?

3 Can you kill a zombie process?
4 Linux process lifecycle.
5 Different stages of a process.
6 What happens internally when you open an application?
7 How does CPU allocate resources?
8 Resources allocation to containers.
9 What does /proc provide?

Boot Process

1 Explain Linux boot process.
2 Explain Linux boot process from power on to login.
3 Role of BIOS.
4 What is MBR?
5 Components of MBR.
6 Linux bootloaders.
7 GRUB vs GRUB2 vs LILO.
8 What is initramfs?
9 First process started by kernel.
10 init vs systemd.
11 Parent process in RHEL6 vs RHEL7+.
12 Kernel panic.
13 Rebuild initramfs.
14 Command to regenerate initramfs.

Services & Systemd

1 What are systemd targets?
2 Manage services using systemctl.
3 Linux run levels.
4 Explain run levels 0-6.
5 Is run level 4 unused?
6 Difference between init and systemd.

Scheduling

1 How do you schedule jobs?

2. cron vs systemd timers.

Logging

1. Monitor logs in real time.
2. tail -f vs journalctl -f.
3. How do you check kernel logs?

Networking (Linux)

1. How do you troubleshoot DNS issues?
2. What are dig and nslookup?
3. Purpose of /etc/hosts.
4. Purpose of /etc/resolv.conf.
5. Check open ports.
6. Capture network packets.
7. Ping troubleshooting.
8. SSH troubleshooting.
9. traceroute.
10. ssh -vvv.
11. Connection timed out.
12. Connection refused.
13. Listening ports.
14. Change SSH port.
15. Verify SSH port.
16. DHCP DORA.
17. OSI model.
18. TCP vs UDP.

Containers (Linux Concepts)

1. Linux namespaces.
2. cgroups.
3. Resource allocation to containers.

Shell Scripting

1 Why Ansible instead of shell scripting?
2 Batch.
3 Shell.
4 Bash scripting.
5 Script to check/update packages on 50 servers.